Name: _____

Class: X Section: A

Subject: MATHEMATICS Date: _____

Chapter/Topic: SOME APPLICATIONS OF TRIGONOMETRY

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MULTIPLE CHOICE QUESTIONS

Q1. The angle of elevation of the top of an object is

(a) the angle between the horizontal through the observer and the upward line of sight (b) the angle between the vertical object and the ground

(c) the angle between the downward line of sight and the horizontal (d) the right angle between the object and the ground

Q2. A vertical pole 8 m high casts a shadow 8√3 m long. The altitude of the Sun is

(a) 60° (b) 30°

(c) 45° (d) cannot be determined

Q3. An 18 m ladder makes an angle of 60° with level ground. The height reached on a vertical wall is

(a) 18√3 m (b) 6√2 m

(c) 9√3 m (d) 3√3 m

Q4. A kite is 30√3 m above level ground. Its taut string makes an angle of 60° with the ground. The string length is

(a) 30 m (b) 45 m

(c) 90 m (d) 60 m

Q5. Meera's eye level is 1.6 m above the ground. From 18.4 m away, she sees the top of a vertical tree at an elevation of 45°. The tree is

(a) 20 m high (b) 18.4 m high

(c) 21.6 m high (d) 36.8 m high

Q6. A 21 m tower casts a shadow when the Sun's altitude is 60°. The shadow length is

(a) $21\sqrt{2}$ m (b) $7\sqrt{3}$ m

(c) $14\sqrt{3}$ m (d) $3\sqrt{3}$ m

Q7. The tops of two vertical poles, 25 m and 13 m high, are joined by a taut wire making 30° with the horizontal. The wire length is

(a) 6 m (b) 12 m

(c) 24 m (d) 48 m

Q8. From the top of a 48 m lighthouse, the angles of depression of two ships on the same side are 30° and 60°. The ships are

(a) $16\sqrt{3}$ m apart (b) $48\sqrt{3}$ m apart

(c) $64\sqrt{3}$ m apart (d) $32\sqrt{3}$ m apart

Q9. Two equal poles stand on opposite sides of a 40 m wide road. From a point between them, their elevations are 45° and 30°. The height of each pole is

(a) $20(\sqrt{3}-1)$ m (b) $20(\sqrt{3}+1)$ m

(c) $40(\sqrt{3}-1)$ m (d) $10(\sqrt{3}-1)$ m

Q10. A 4 m statue stands on a pedestal. From a point on the ground, the elevations of the top of the pedestal and the top of the statue are 45° and 60°. The pedestal height is

(a) $2(\sqrt{3}-1)$ m (b) $2(\sqrt{3}+1)$ m

(c) $4(\sqrt{3}+1)$ m (d) $\sqrt{3}+1$ m

Q11. From the roof of a 7 m building, the elevation of the top of a tower is 45° and the depression of its foot is 30°. The tower height is

(a) $7(\sqrt{3}-1)$ m (b) $7\sqrt{3}$ m

(c) $7(\sqrt{3}+1)$ m (d) $14(\sqrt{3}+\sqrt{2})$ m

Q12. A tree breaks so that its top touches level ground 15√3 m from the foot, and the broken part makes 30° with the ground. The original height was

(a) 15 m (b) 30 m

(c) 60 m (d) 45 m

Q13. Two aircraft are vertically one above the other. The upper aircraft is 3600 m high. From a ground point their elevations are 60° and 45°, respectively. Their vertical separation is

(a) $(3600\sqrt{3}-1200)$ m (b) $(3600+1200\sqrt{3})$ m

(c) $1200\sqrt{3}$ m (d) $(2400+1200\sqrt{3})$ m

Q14. A 6 m flagstaff stands on a tower. From a ground point, the elevations of the top of the tower and the top of the flagstaff are 45° and 60°. The tower height is

(a) $3(\sqrt{3}-1)$ m (b) $3(\sqrt{3}+1)$ m

(c) $6(\sqrt{3}+1)$ m (d) 6√3 m

Q15. The elevations of the top of an 18√3 m tower from two points on opposite sides are 30° and 60°. The distance between the points is

(a) 36 m (b) 54 m

(c) 72 m (d) 108 m

Q16. From a window 12 m above the ground, the depression of a tower's foot is 45° and the elevation of its top is 30°. The tower height is

(a) $12 + 4\sqrt{3}$ m (b) $12\sqrt{3}$ m

(c) $24 + 4\sqrt{3}$ m (d) $12 + 4\sqrt{3}$ m

Q17. From a point 20 m above a lake, a cloud is seen at 30° elevation and its reflection at 60° depression. The distance from the observer to the cloud is

(a) 40 m (b) 20 m

(c) 60 m (d) 80 m

Q18. A tower stands on one bank of a canal. From the directly opposite point on the other bank its elevation is 60°. From a point 30 m farther from the tower on the same line it is 30°. The canal is

(a) 30 m wide (b) 15 m wide

(c) 45 m wide (d) 60 m wide

Q19. A balloon moves horizontally 90 m above an observer's eye level. Its elevation changes from 60° to 30°. The horizontal distance travelled is

(a) $30\sqrt{3}$ m (b) $90\sqrt{3}$ m

(c) $60\sqrt{3}$ m (d) $120\sqrt{3}$ m

Q20. From two points 20 m apart on the same side of a tower, the nearer point has elevation 60° and the farther point 45°. The tower height is

(a) $(30 - 10\sqrt{3})$ m (b) $(20 + 10\sqrt{3})$ m

(c) $(60 + 10\sqrt{3})$ m (d) $(30 + 10\sqrt{3})$ m

ASSERTION–REASON QUESTIONS

Direction: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as
(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
(c) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true.

Q1. Assertion (A): A 10-m vertical pole casting a 10-m shadow indicates a solar elevation of 45°.
Reason (R): For a vertical pole, $\tan\theta = \text{height/shadow}$, and $\tan 45° = 1$.

Q2. Assertion (A): A 12-m ladder inclined at 60° to level ground reaches a height of 6√3 m.
Reason (R): The angle of elevation is the angle between the horizontal and the upward line of sight.

Q3. Assertion (A): The angle of depression of a car from a tower equals the car's angle of elevation to the tower top.
Reason (R): These two angles are vertically opposite angles.

Q4. Assertion (A): In a heights-and-distances right-triangle model, an angle of elevation may be 120°.
Reason (R): The angle of elevation is measured upward from the horizontal to the line of sight and is acute in this model.

Q5. Assertion (A): A 24 m tower viewed at an elevation of 60° is 8√3 m from the observer.
Reason (R): For a vertical tower, tan 60° = height/horizontal distance = √3.

Q6. Assertion (A): A flagstaff stands on a tower. If the elevations of the tower top and flagstaff top are 45° and 60°, the tower height equals the horizontal distance from the observer.
Reason (R): tan 60° = √3/2.

Q7. Assertion (A): A ship sees at a depression of 30° from a 30 m lighthouse is 30√3 m from its foot.
Reason (R): In a right triangle, cos 90° = opposite side/hypotenuse.

Q8. Assertion (A): The angle of depression from a tower top to a point on level ground can be −45°.
Reason (R): An angle of depression is measured downward from the horizontal through the observer.

Q9. Assertion (A): Two equal poles stand 20 m apart. From their midpoint, each top has elevation 45°, so each pole is 10 m high.
Reason (R): For either right triangle, tan 45° = height/10 = 1.

Q10. Assertion (A): When a broken tree touches the ground, its fallen upper part is the hypotenuse of the right triangle formed.
Reason (R): The angle of depression equals the corresponding angle of elevation because horizontal lines are parallel.

Q11. Assertion (A): For a fixed vertical tower, an observer's angle of elevation increases as the observer moves closer on level ground.
Reason (R): tan θ increases as θ increases from 30° to 60°.

Q12. Assertion (A): A 20 m taut rope inclined at 30° to the ground reaches a vertical height of −10 m.
Reason (R): For this right triangle, $\sin 30° = \text{vertical height}/\text{rope length} = 1/2$.

Q13. Assertion (A): A tower is seen at 30°. After the observer moves 20 m towards it, the elevation is 60°, the tower is $10\sqrt{3}$ m high.
Reason (R): For a vertical object, $\tan\theta = \text{height}/\text{horizontal distance}$ at each observation point.

Q14. Assertion (A): A 1.5 m tall observer standing 28.5 m from a chimney sees its top at 45°, so the chimney is 30 m high.
Reason (R): $\sin\theta\,\sec\theta = \cos\theta$ = 1 for an acute angle θ.

Q15. Assertion (A): Of two ships on the same side of a lighthouse, the nearer ship subtends the larger angle of depression.
Reason (R): An angle of depression is measured upward from the horizontal through the observer.

Q16. Assertion (A): For a fixed tower height, the horizontal distance is greater at an elevation of 60° than at 45°.
Reason (R): For tower height h and elevation θ, horizontal distance $d = h\cot\theta$.

Q17. Assertion (A): From a point 20 m above a lake, a cloud at 30° elevation and its reflection at 60° depression place the cloud 40 m above the lake.
Reason (R): A reflection is as far below the lake surface as the cloud is above it, and tan θ = vertical change/horizontal distance.

Q18. Assertion (A): The top of a 15 m tower viewed from 5√3 m away has an elevation of 60°.
Reason (R): The hypotenuse is the longest side of a right triangle.

Q19. Assertion (A): At a fixed elevation of 45°, doubling the horizontal distance doubles the required vertical height.
Reason (R): tan 45° = √3.

Q20. Assertion (A): Two equal poles are seen from a point between them at 60° and 30°. The nearer and farther horizontal distances are in the ratio 1 : 3.
Reason (R): Since the poles have equal heights, the point must be equidistant from them.